functions designed to create or remove elements that you can use to generate content. We'll cover selections and data-binding in detail in chapter 2. Selections might not include any data-binding, and won't for most of the examples in this chapter, but the inclusion allows the powerful information visualization techniques of D3. You can make a selection on any elements in a web page, including items in a list, circles, or even regions on a map of Africa. Just as the elements can take a number of shapes, the data associated with those elements (where applicable) can take many forms.

1.2.3 D3 is about deriving the appearance of web page elements from bound data

After you have a selection, you can then use D3 to modify the appearance of web page elements to reflect differences in the data. You may want to make the length of a line equal to the value of the data, or change the color to a particular color that corresponds to a class of data. You may want to hide or show elements as they correspond to a user's navigation of a dataset. As you can see in figure 1.9, after the page has loaded, you use D3 to select elements and bind data for the purpose of creating, removing, or changing DOM elements. You continue to use this process in response to user interaction.

You modify the appearance of elements by using selections to reference the data bound to an element in a selection. D3 iterates through the elements in your selection and performs the same action using the bound data, which results in different graphical

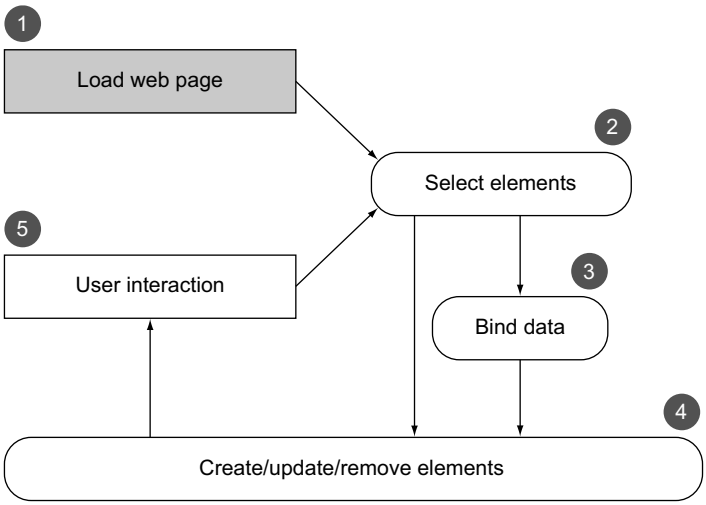


Figure 1.9 A page utilizing D3 is typically built in such a way that the page loads with styles, data, and content as defined in traditional HTML development ❶ with its initial display using D3 selections of HTML elements ❷, either with data-binding ❸ or without it, to modify the structure and appearance of the page ❹. The changes in structure prompt user interaction ❺, which causes new selections with and without data-binding to further alter the page. Step 1 is shown differently because it only happens once (when you load the page), whereas every other step may happen multiple times, depending on user interaction.